Quantum Minesweeper: Safe Routes

Developer: Oļģerts Osipovs

Core Concept

In *Quantum Minesweeper: Safe Routes*, the player must establish a safe evacuation corridor across a city-grid under incomplete information. Each cell does not start as a fixed "safe" or "dangerous" tile; instead, it exists in an uncertain state described by a risk likelihood. Only when the player decides to check a tile (or steps onto an unknown tile) does the game resolve that uncertainty into a concrete outcome.

The intended experience is decision-making under real constraints: caution versus speed, certainty versus cost. The player is constantly choosing between spending limited checks to gain reliable knowledge or moving faster and accepting the consequences of uncertainty.

Thematic Focus (Social Good)

The theme is public safety and humanitarian logistics: building safe routes for evacuation/accessibility when resources are limited and the situation is uncertain. The social-good connection is integrated into gameplay rather than added as narrative decoration: the game models how real-world verification (inspections, checks, scanning areas) reduces risk but also comes with social costs (delays, fatigue, loss of trust).

This creates a meaningful ethical tension: the "best" strategy is not simply to check everything, but to find a sustainable balance between safety, speed, and the social impact of over-monitoring.

Quantum Dimension

The game implements quantum ideas at a mechanical / state-based level:

- Superposition: before interaction, a tile is not a determined safe/danger state; it is represented by a probability of danger (uncertainty is fundamental until resolved).
- Measurement: the SCAN action acts as measurement, revealing the tile's realized state.
- Collapse: after measurement (or stepping onto an unscanned tile), the tile collapses into a definite SAFE or DANGER outcome.
- Observer Effect: measurements are not free—each scan reduces a Trust metric, reflecting that observation/verification changes the system (social response, compliance, fatigue).
- Entanglement (simple form): some tiles are linked in pairs; measuring one immediately affects or resolves the other, showing that parts of the system can be correlated rather than independent.

How it goes beyond a purely classical probabilistic model: the player is not only managing random risk. The act of observation carries an explicit cost that changes future outcomes (Trust), and entangled tiles break the assumption of independent events. The core challenge becomes optimizing a coupled system of uncertainty, cost of knowledge, and social impact.

Game Mechanics and Rules

Structure: grid-based play with two modes: SCAN and STEP.

Player actions:

1. SCAN (measure a tile): click a tile to measure it.
    - Result: the tile collapses to SAFE or DANGER.
    - Cost: consumes limited scans and reduces Trust.
2. STEP (move the evac marker): move one tile (4-directional).
    - If stepping onto an unmeasured tile, it collapses on contact and reduces Trust more strongly.
    - If it collapses to DANGER, casualties increase.

Constraints / limitations:

- limited number of scans,
- Trust decreases with scans and risky movement,
- a maximum allowed casualty count.

System responses:

- tile states update after each scan/step,
- entangled pairs may collapse together or influence each other when one is measured.

Setting

The setting is a city after a disruption (e.g., infrastructure failure or emergency conditions). The player acts as a coordinator responsible for guiding people to safety. The narrative supports the theme by framing “measurement” as real-world verification (inspection/scanning) with real costs, and **Trust** as a proxy for social stability and cooperation.

Winning and Losing Conditions

Win: reach the Exit from Start while keeping Trust > 0 and staying within the casualty limit.
Lose:

- Trust reaches zero (social trust collapses), or
- casualties exceed the allowed limit.

Optional scoring/outcomes can add multiple endings (excellent route vs. barely acceptable route) based on remaining Trust and casualties.

Platform and Tools

- **Platform:** Pygame
- **Tools / language:** Python + Pygame